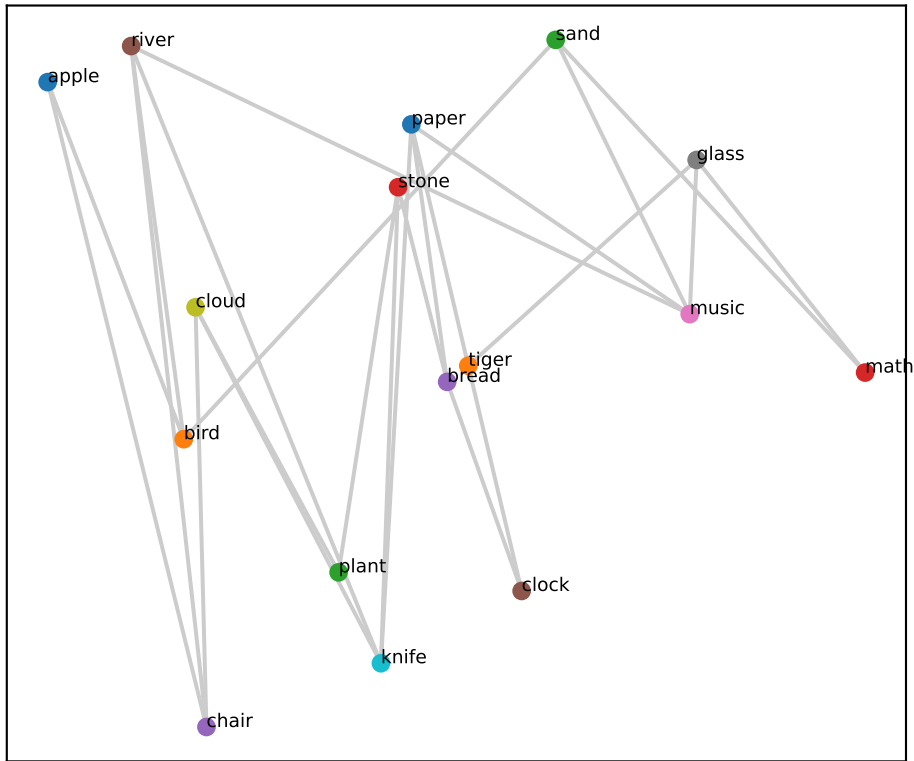
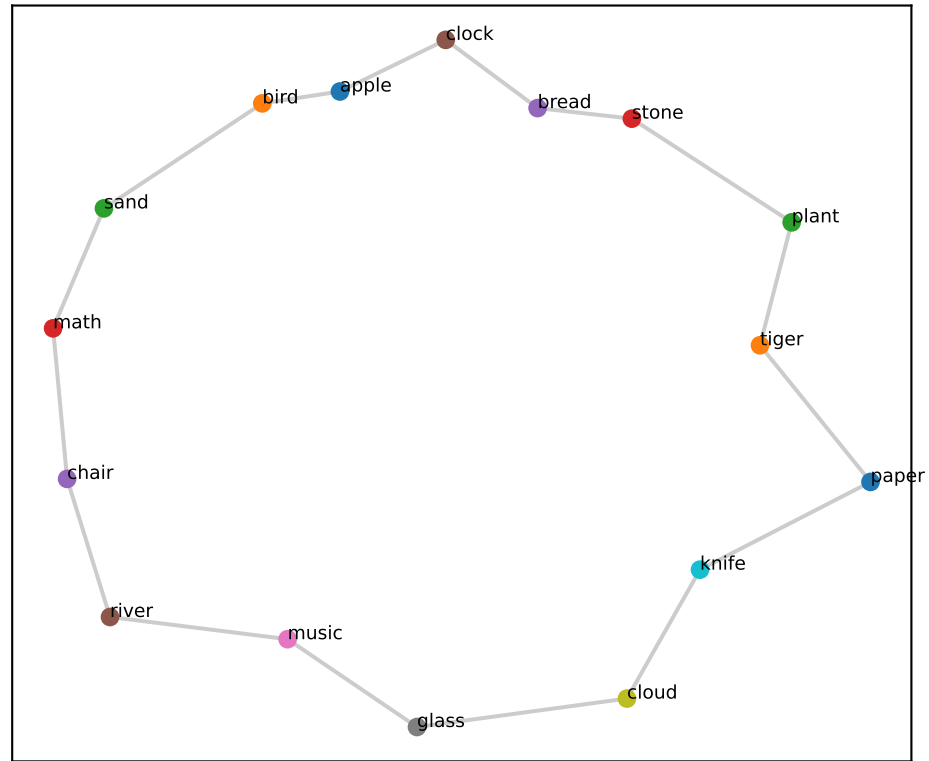


Llama Top: PCA top-2 (variance) | Bottom: best-2D (top-6 PCs -> coords); null = label-permutation through the fit

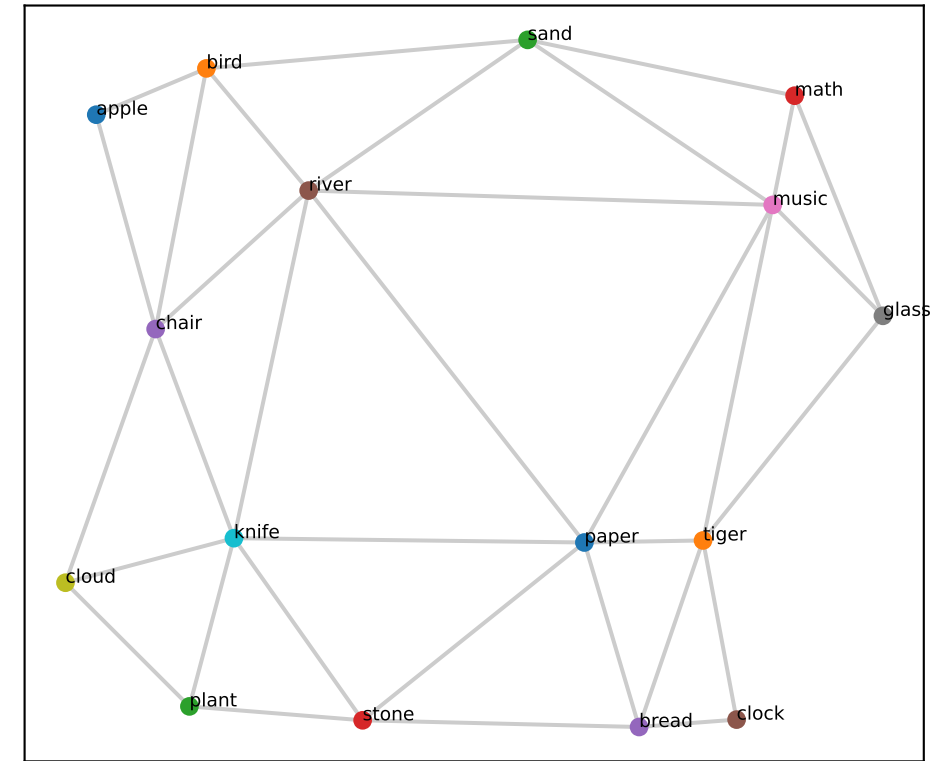
square_grid PCA-2D (RSA 0.31)



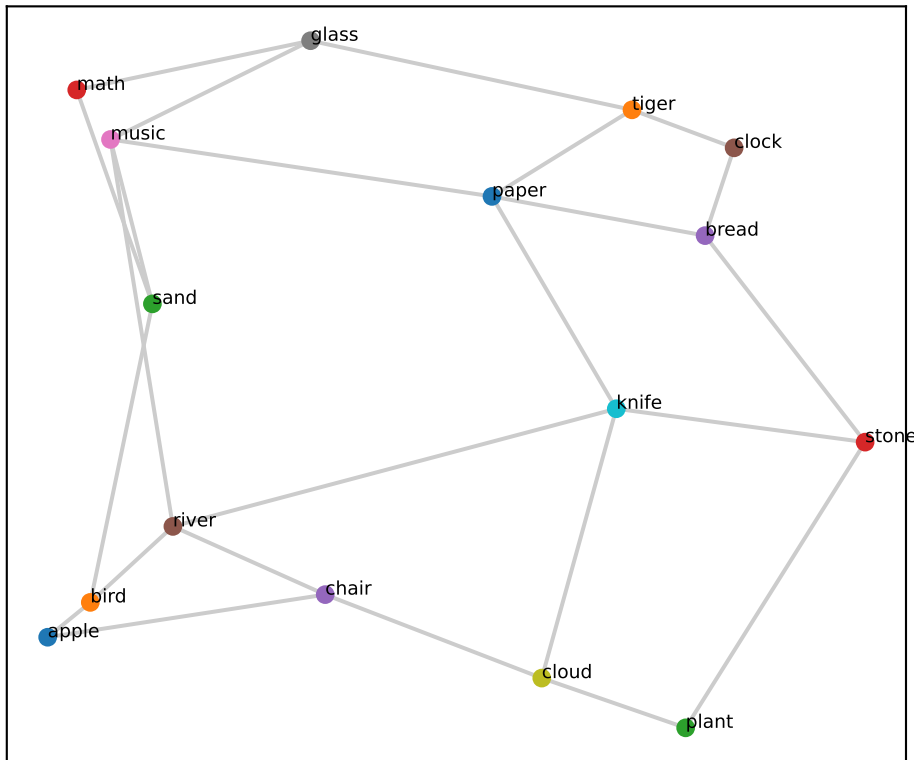
ring PCA-2D (RSA 0.92)



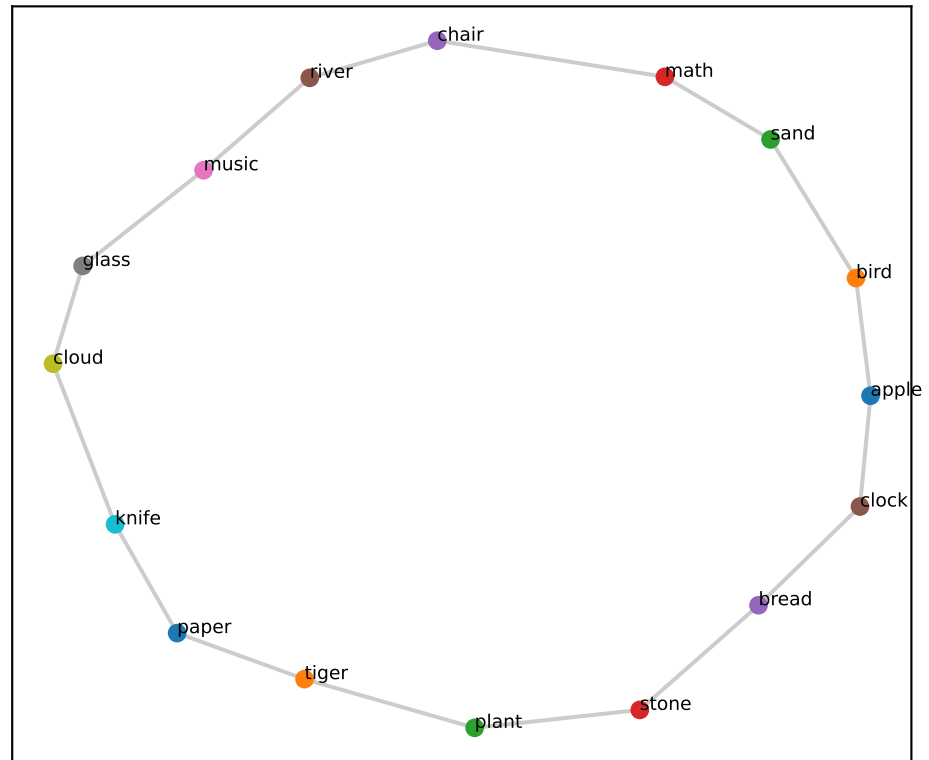
hex PCA-2D (RSA 0.85)



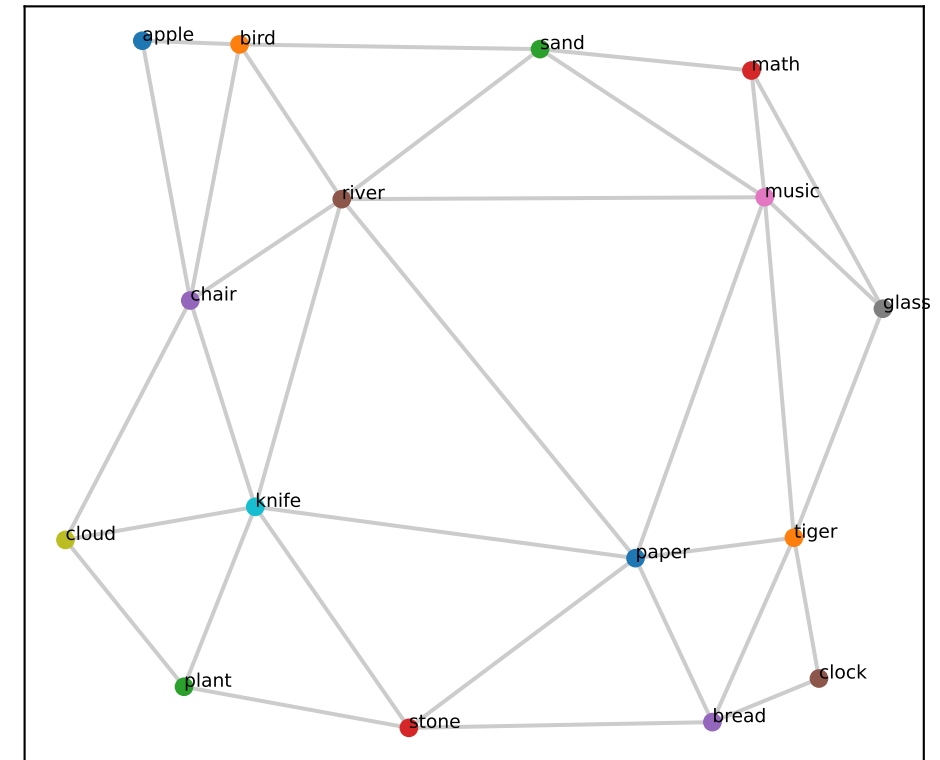
square_grid best-2D (RSA 0.81 | null95 0.48, p<.001)



ring best-2D (RSA 0.98 | null95 0.50, p<.001)

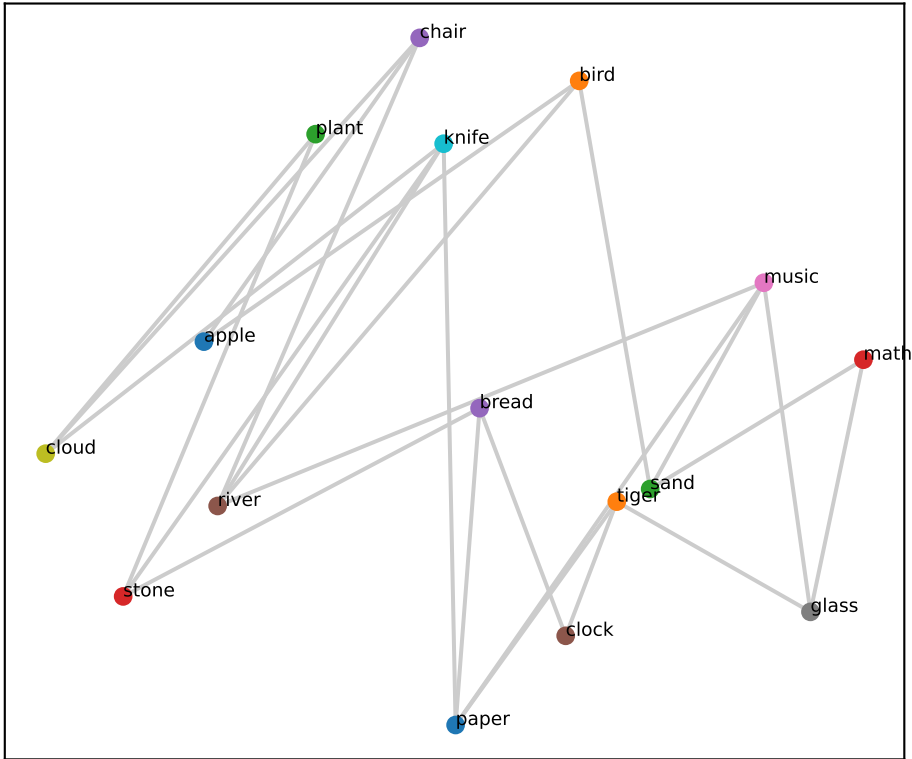


hex best-2D (RSA 0.88 | null95 0.48, p<.001)

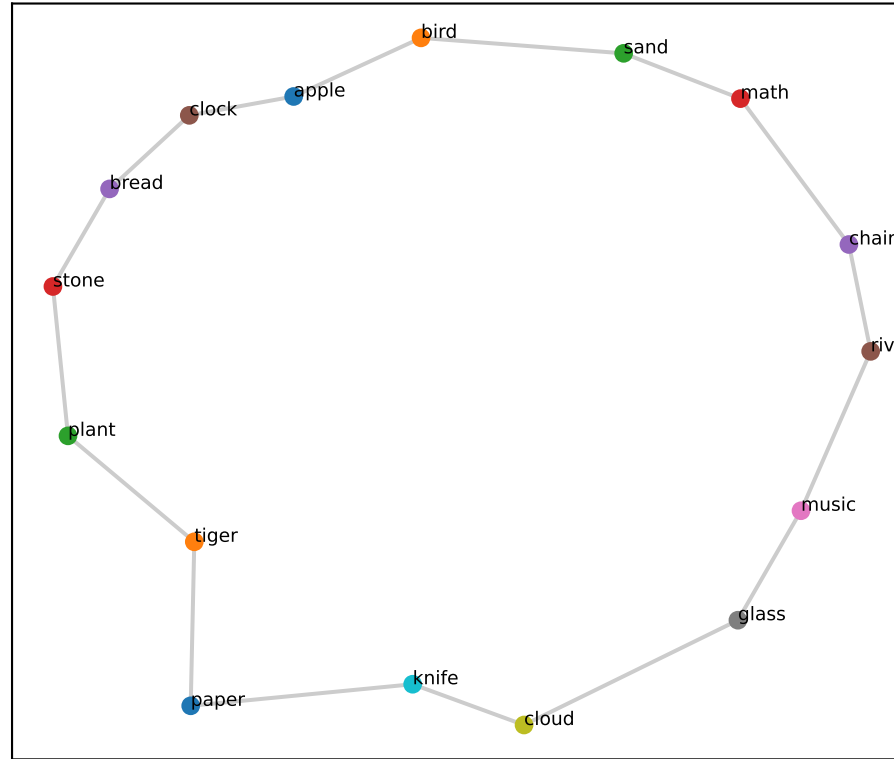


Gemma Top: PCA top-2 (variance) | Bottom: best-2D (top-6 PCs -> coords); null = label-permutation through the fit

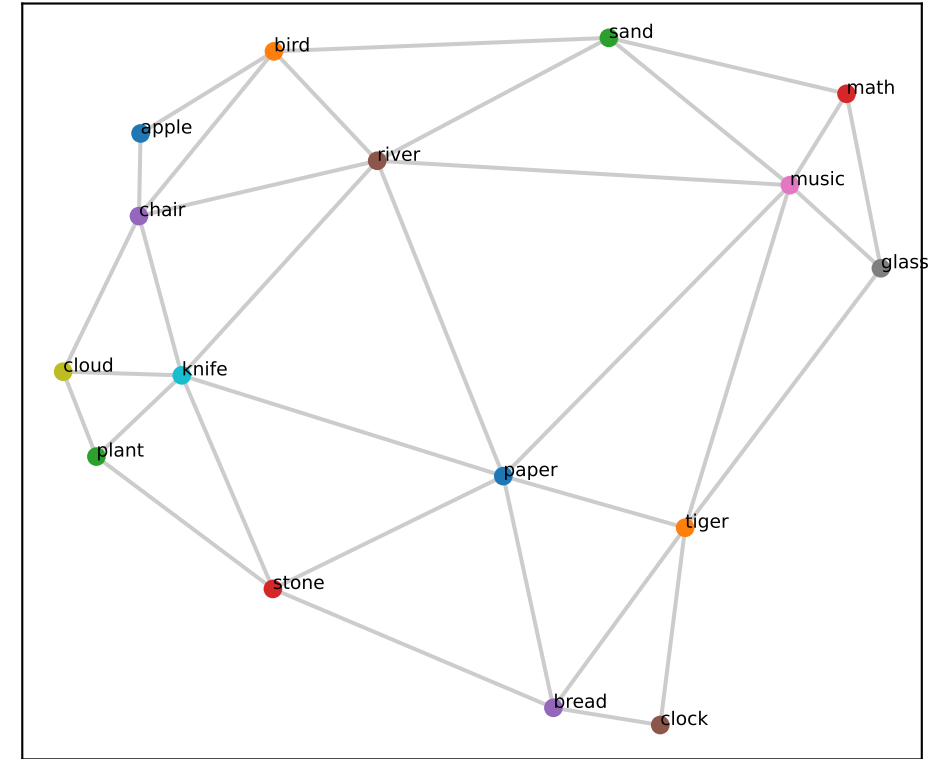
square_grid PCA-2D (RSA 0.36)



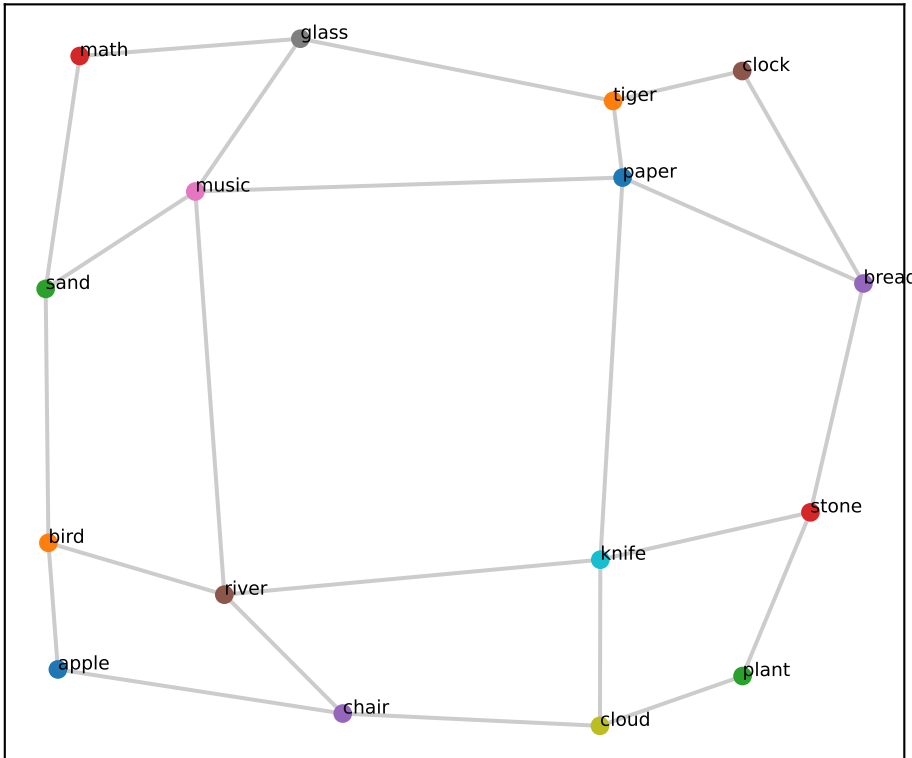
ring PCA-2D (RSA 0.95)



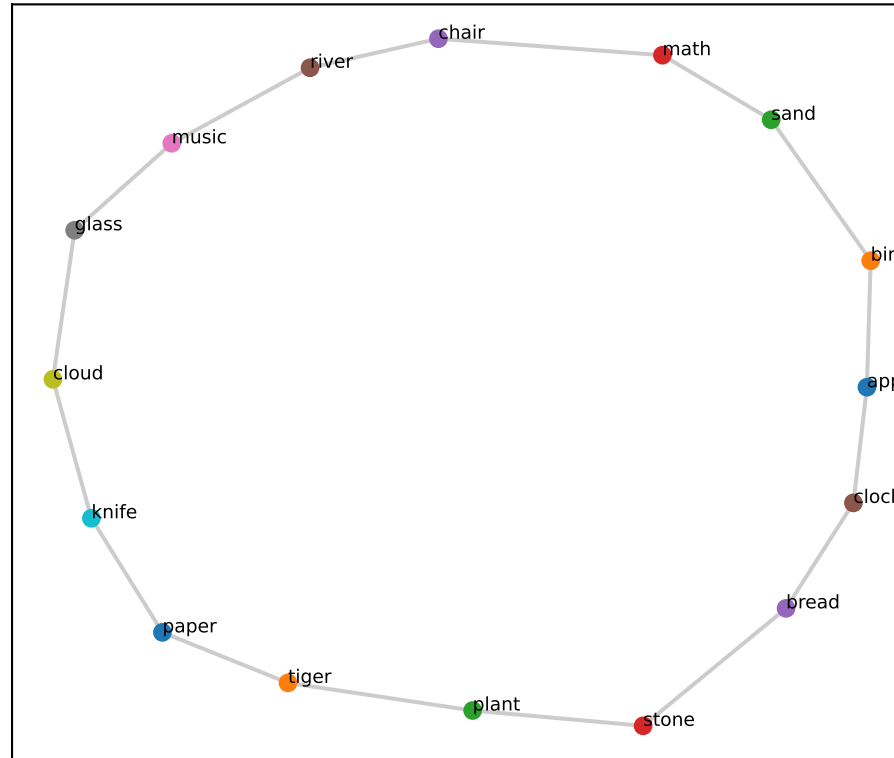
hex PCA-2D (RSA 0.86)



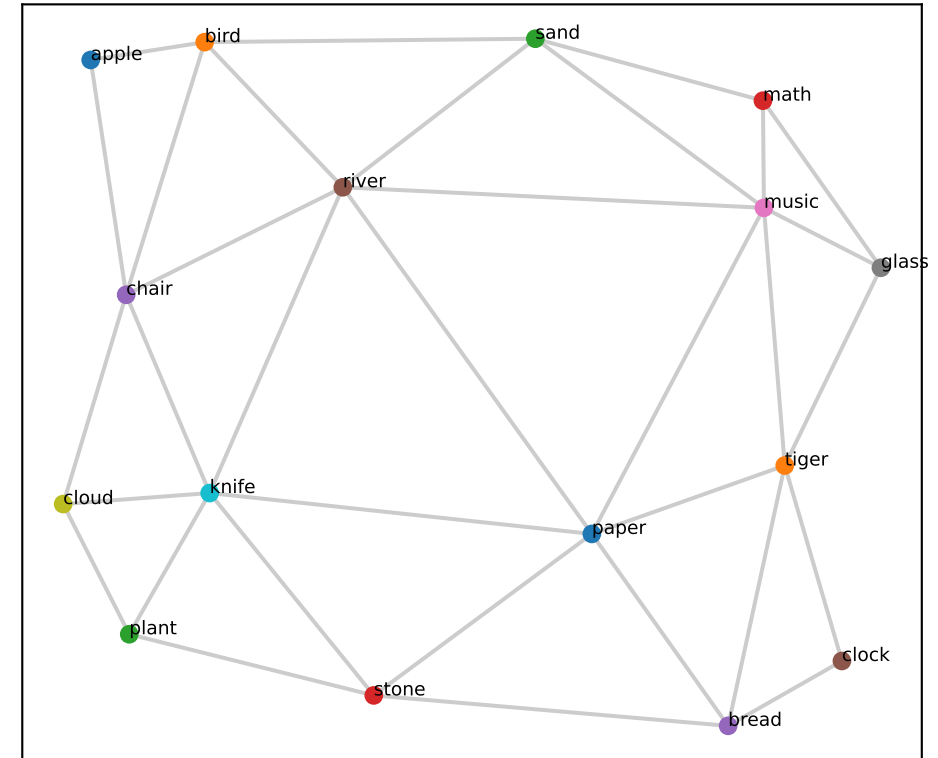
square_grid best-2D (RSA 0.89 | null95 0.47, p<.001)



ring best-2D (RSA 0.98 | null95 0.49, p<.001)

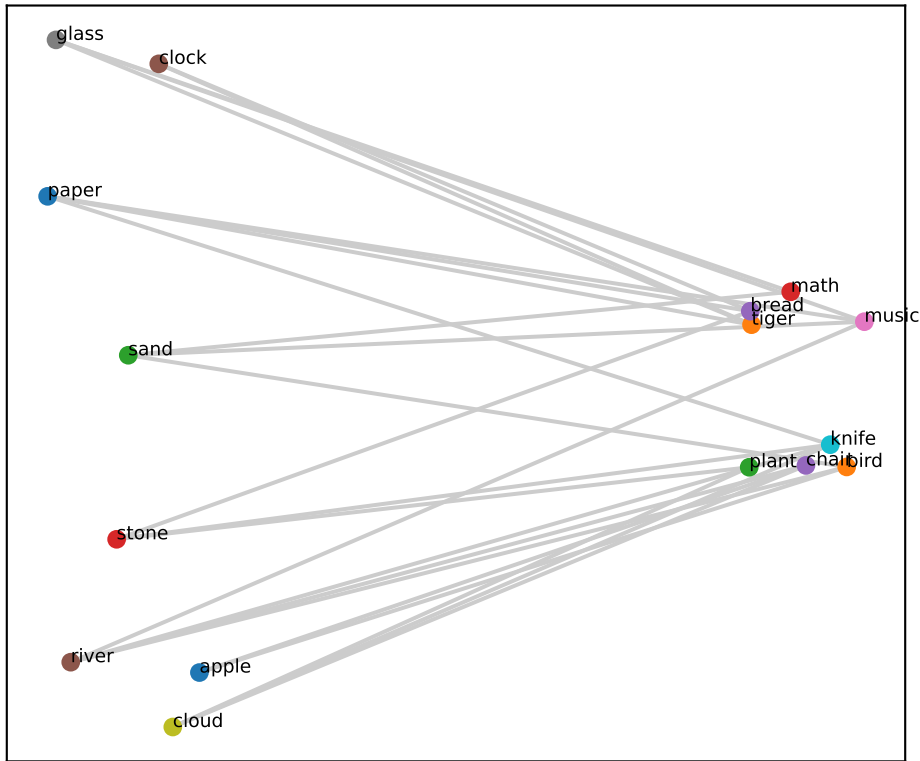


hex best-2D (RSA 0.89 | null95 0.48, p<.001)

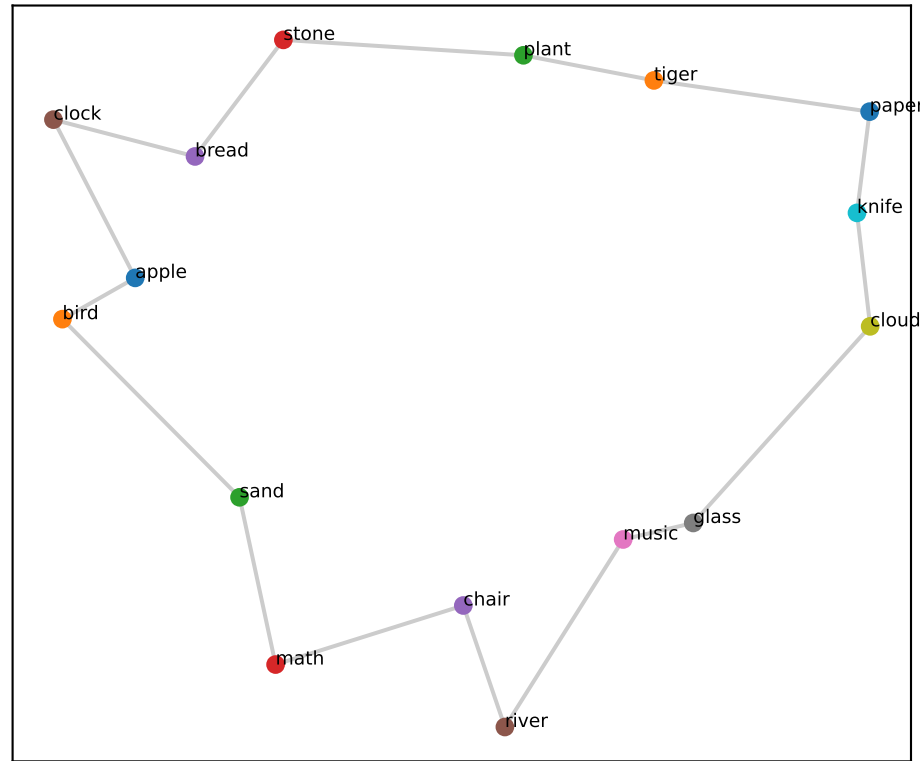


Qwen Top: PCA top-2 (variance) | Bottom: best-2D (top-6 PCs -> coords); null = label-permutation through the fit

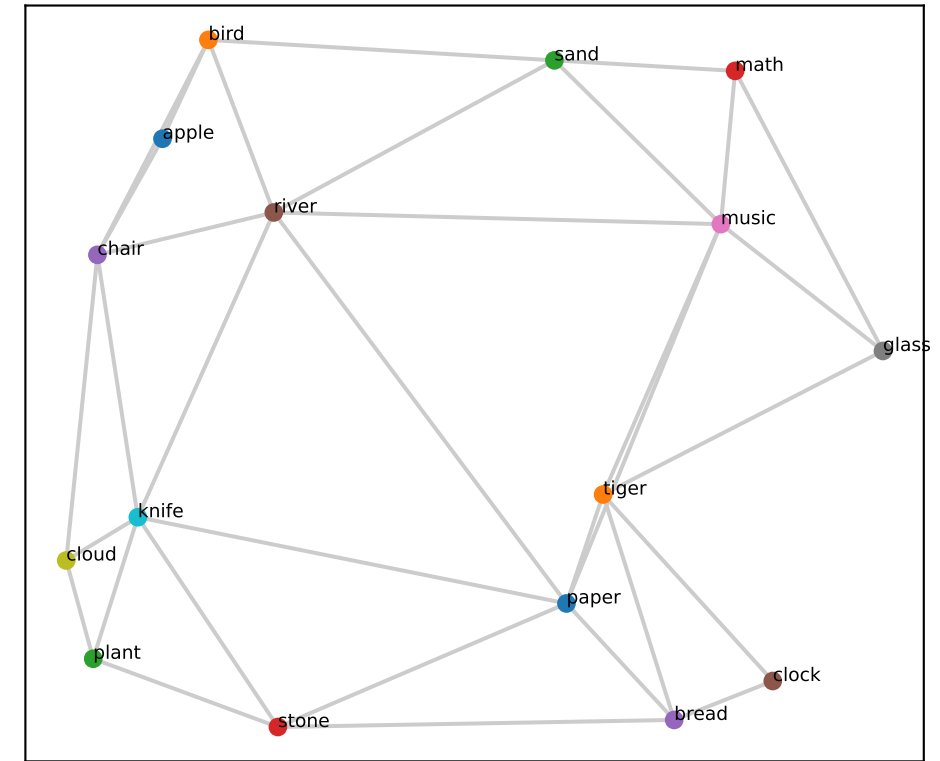
square_grid PCA-2D (RSA 0.08)



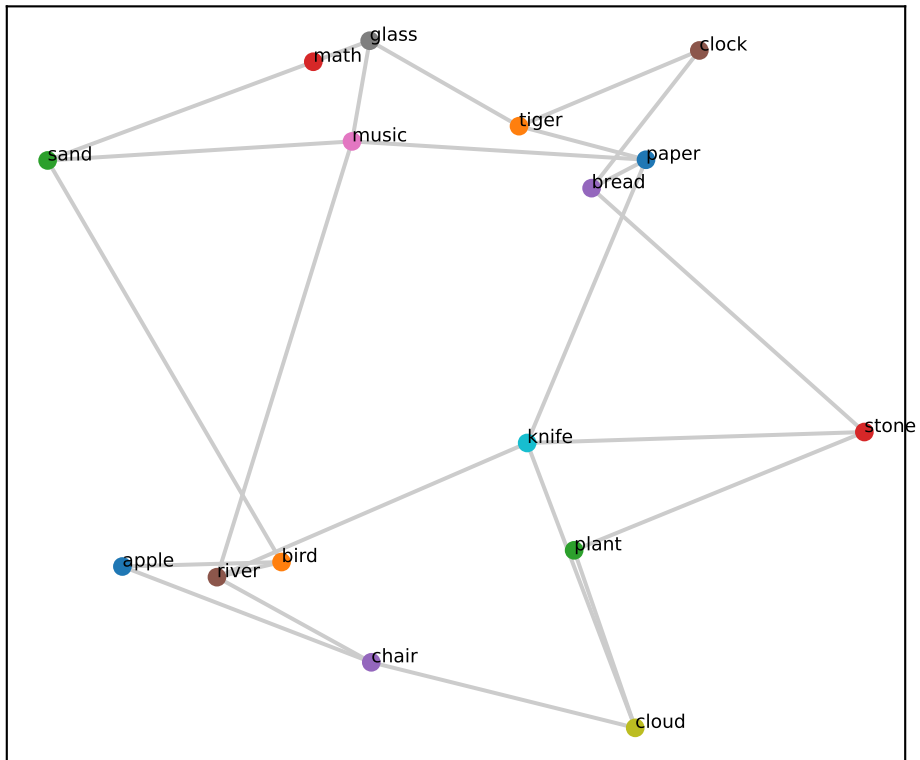
ring PCA-2D (RSA 0.89)



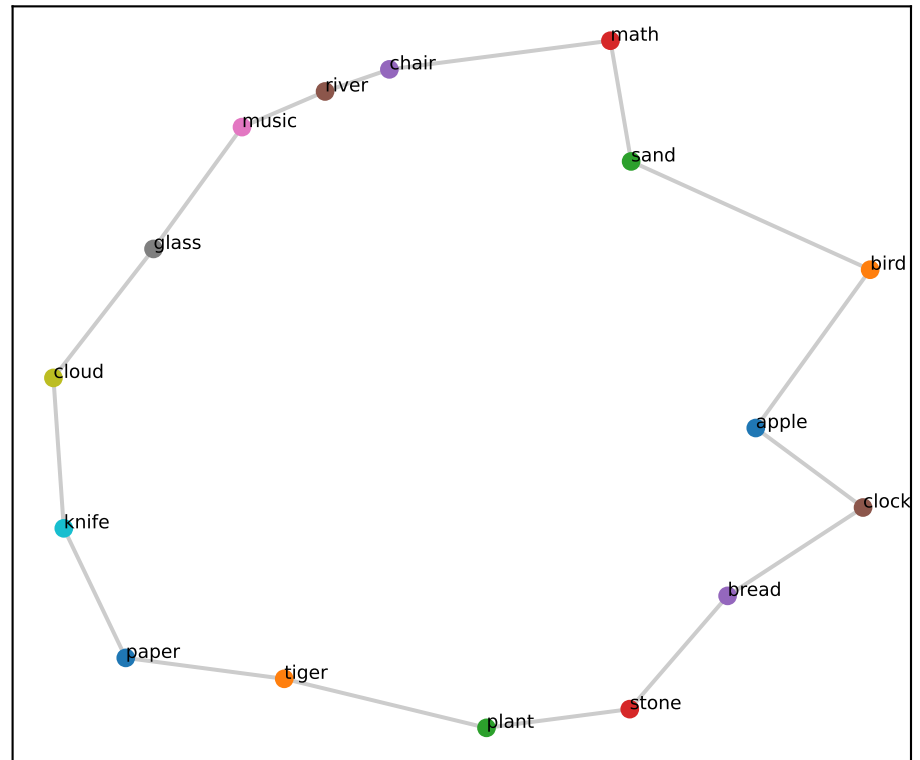
hex PCA-2D (RSA 0.82)



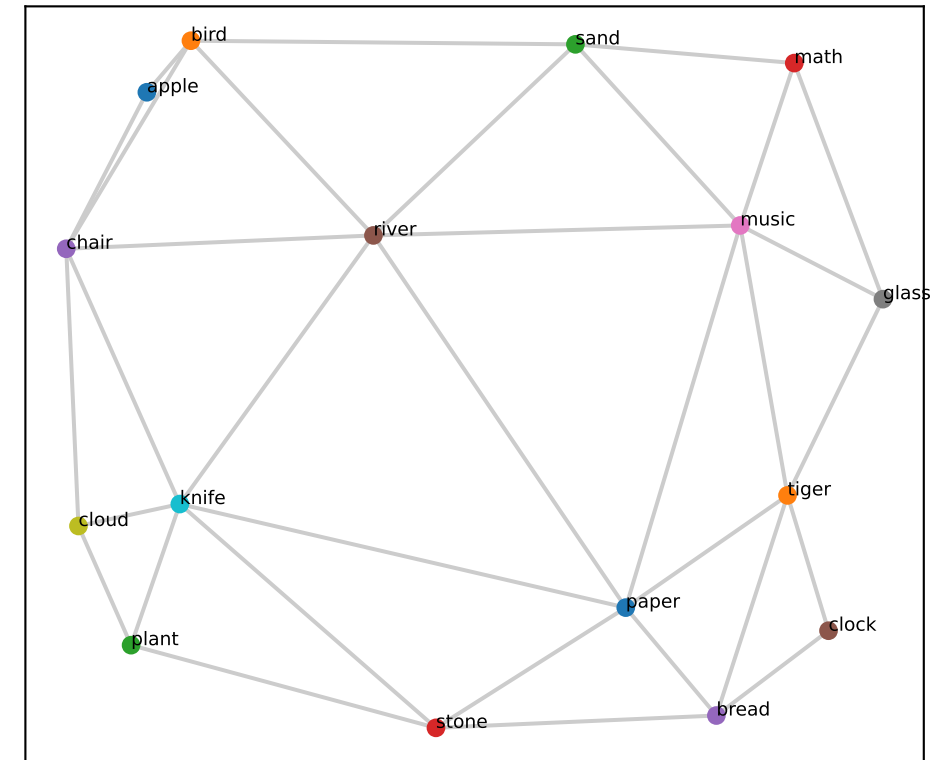
square_grid best-2D (RSA 0.72 | null95 0.46, $p < .001$)



ring best-2D (RSA 0.94 | null95 0.49, $p < .001$)

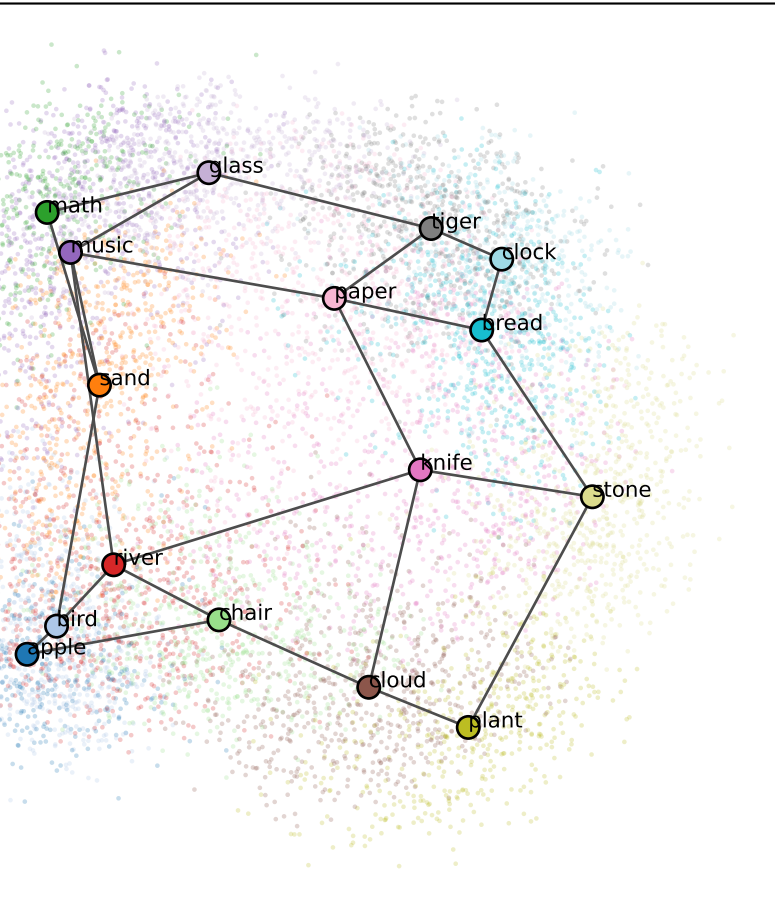


hex best-2D (RSA 0.88 | null95 0.48, $p < .001$)

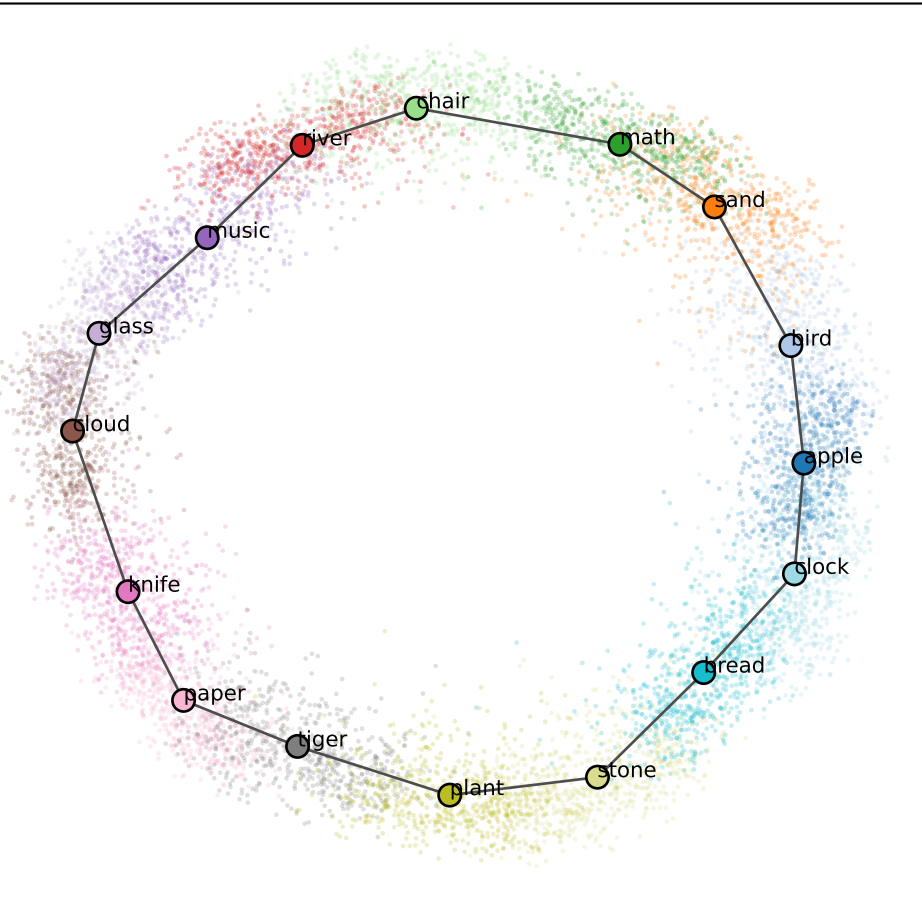


Llama — every occurrence on the two best-2D axes (dots=occurrences, markers=node means)

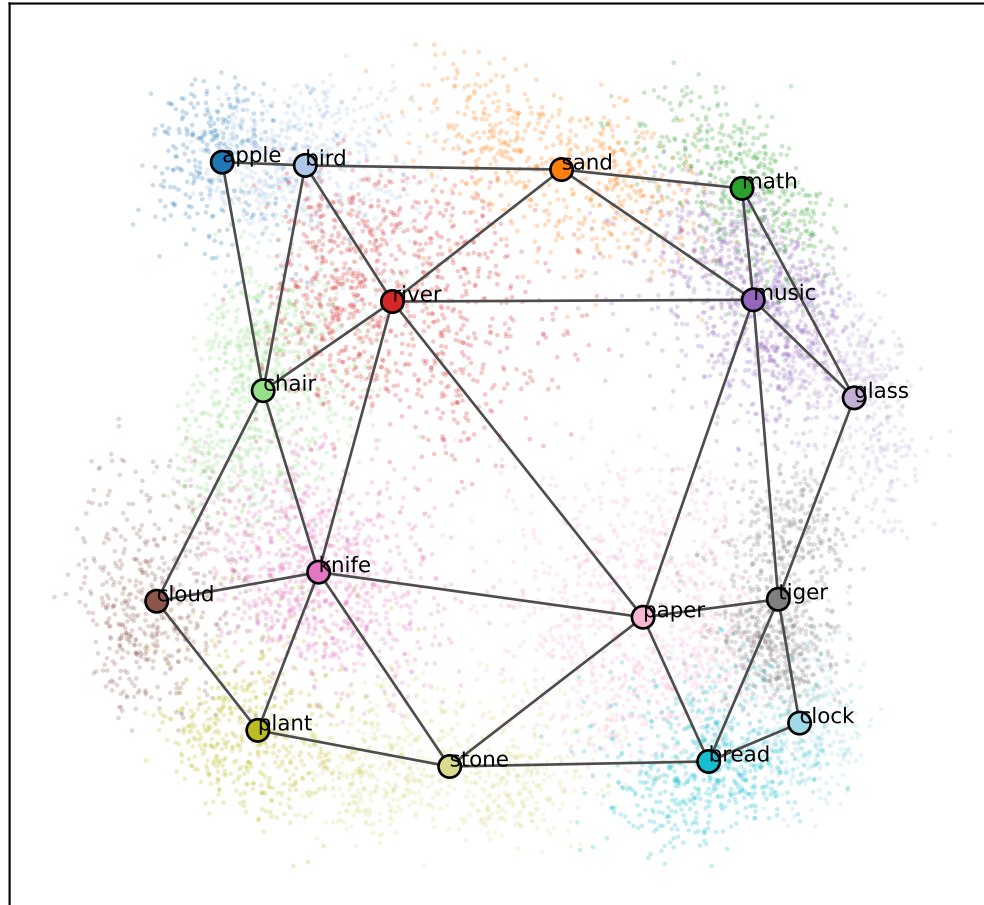
square_grid best-2D (RSA 0.81, $p < .001$)



ring best-2D (RSA 0.98, $p < .001$)

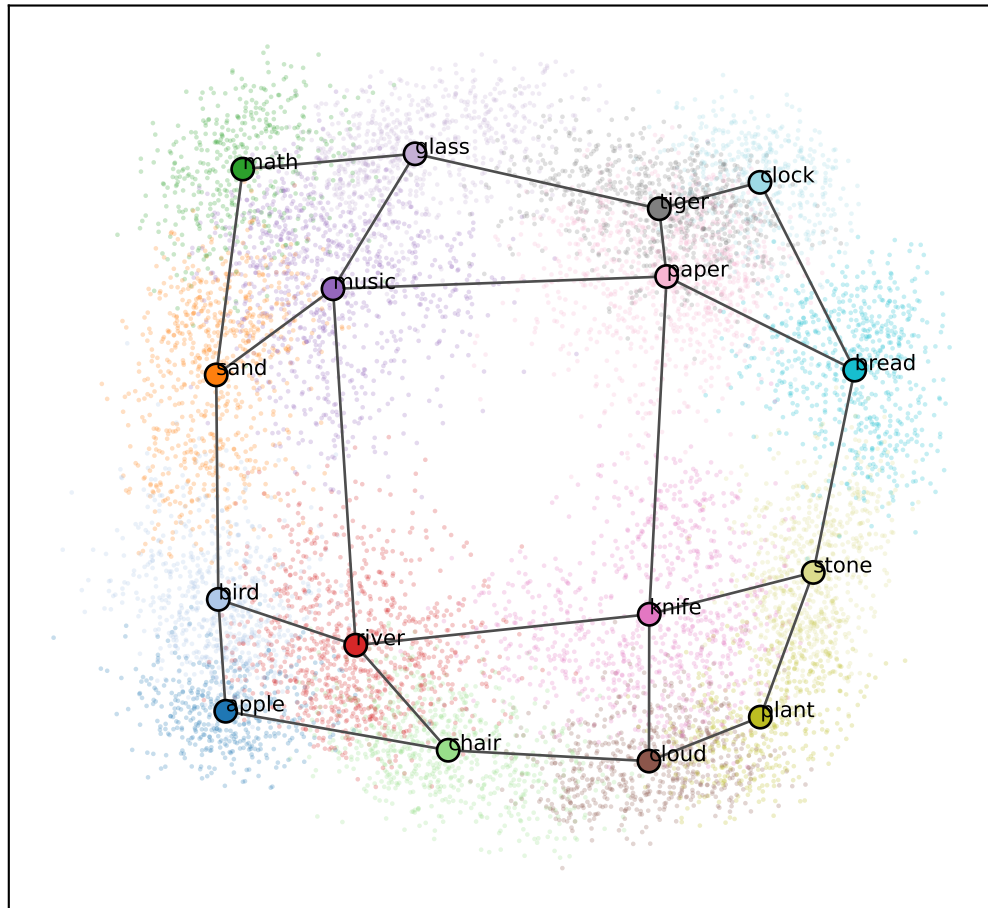


hex best-2D (RSA 0.88, $p < .001$)

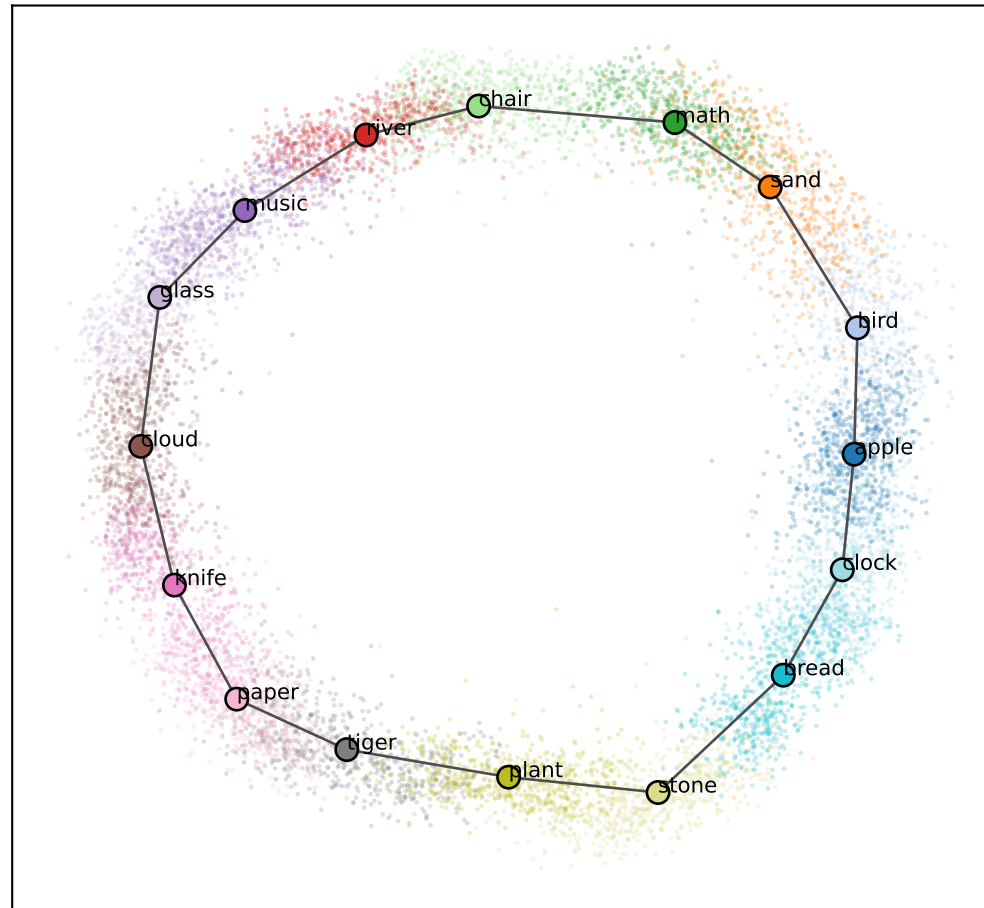


Gemma — every occurrence on the two best-2D axes (dots=occurrences, markers=node means)

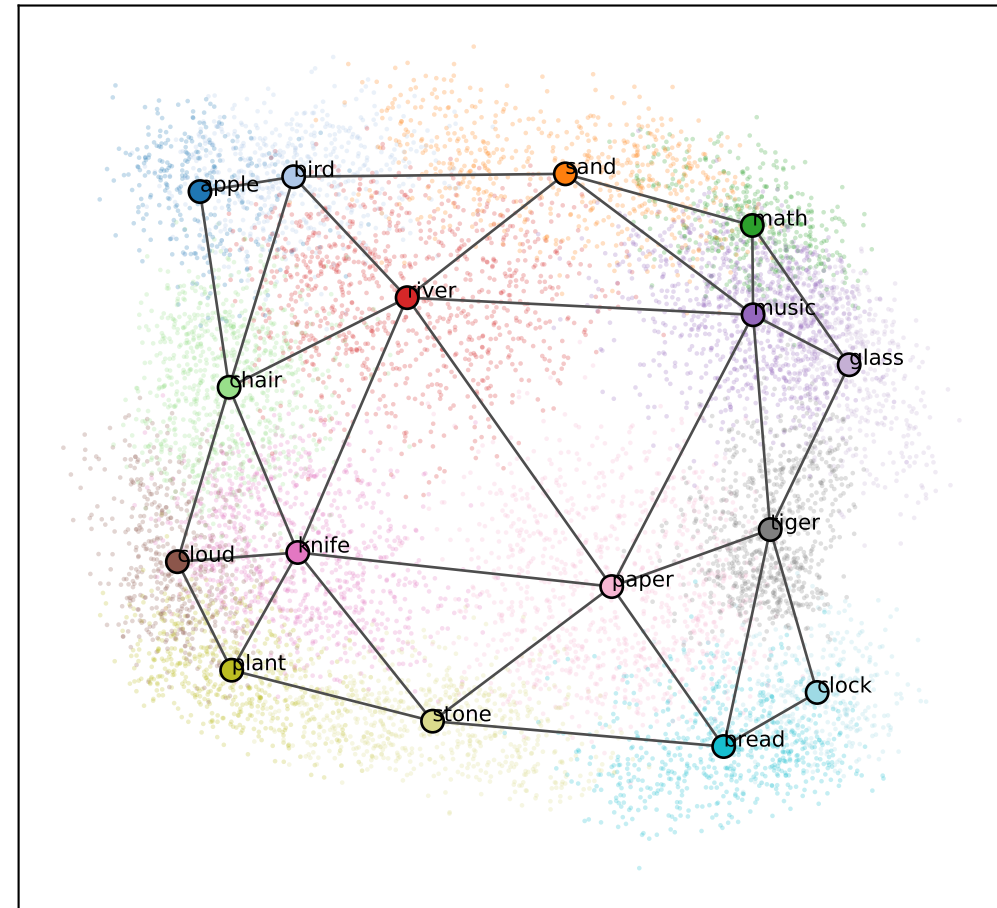
square_grid best-2D (RSA 0.89, $p < .001$)



ring best-2D (RSA 0.98, $p < .001$)

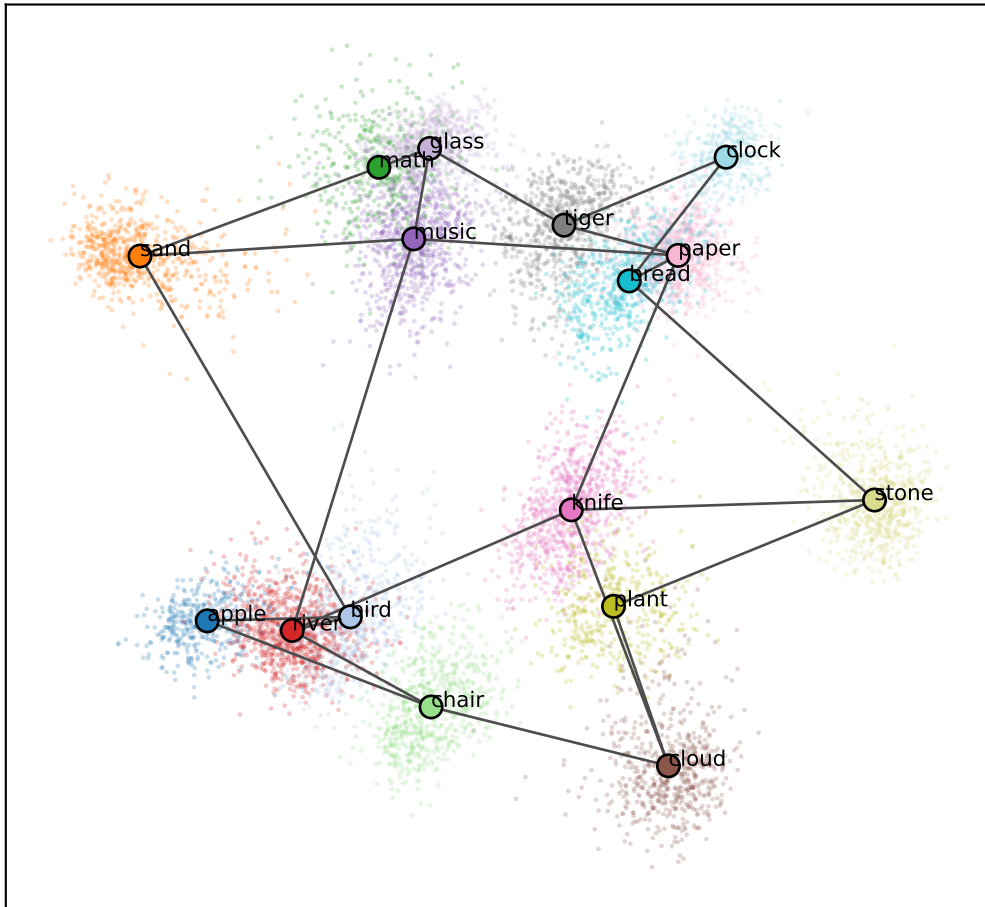


hex best-2D (RSA 0.89, $p < .001$)

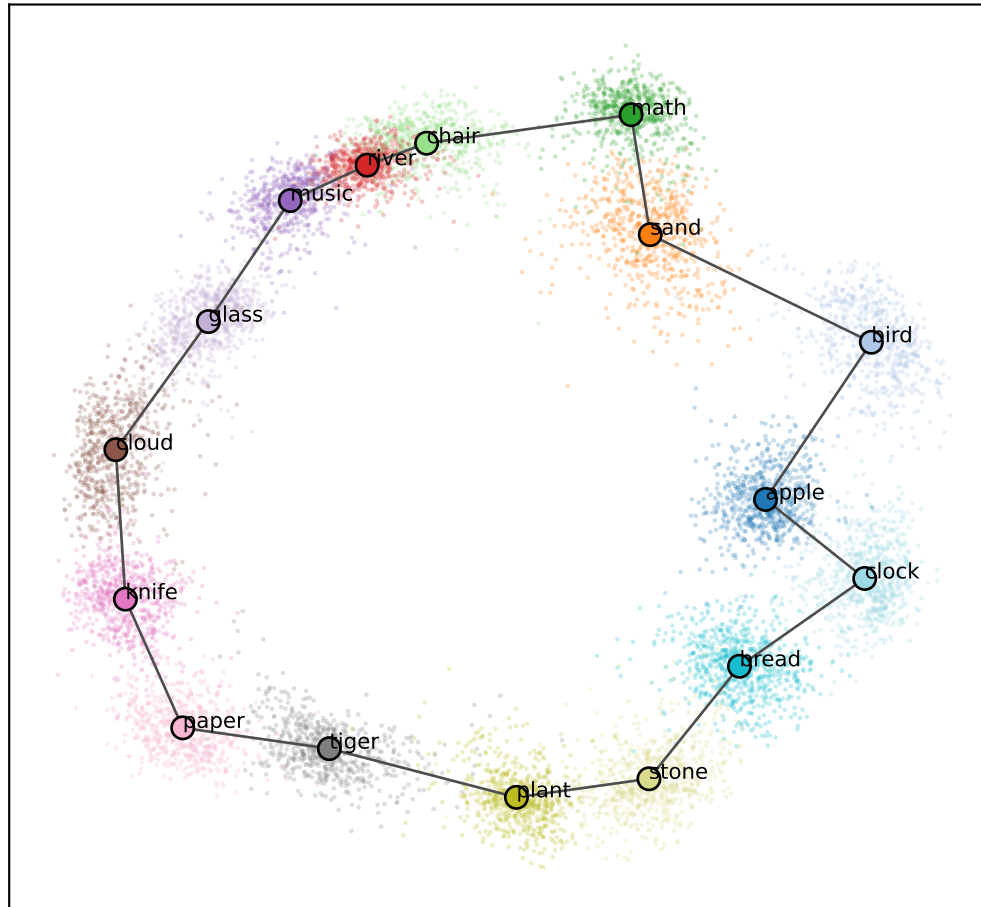


Qwen — every occurrence on the two best-2D axes (dots=occurrences, markers=node means)

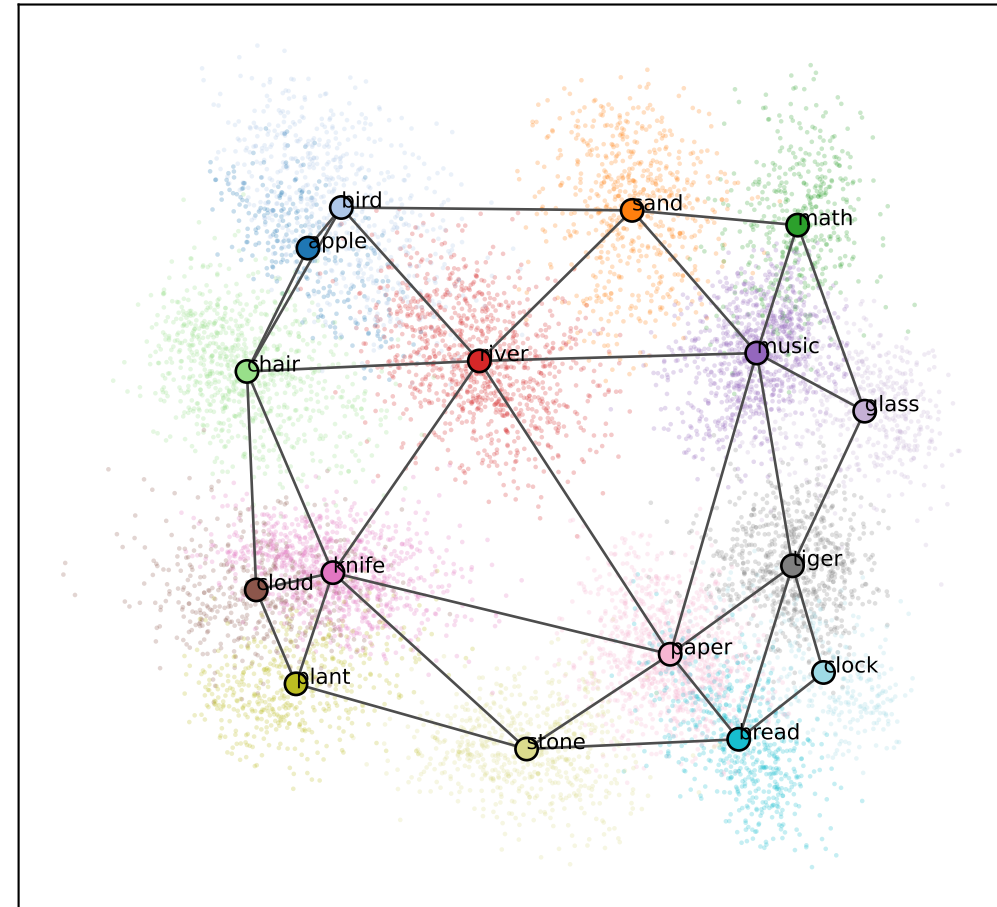
square_grid best-2D (RSA 0.72, $p < .001$)



ring best-2D (RSA 0.94, $p < .001$)

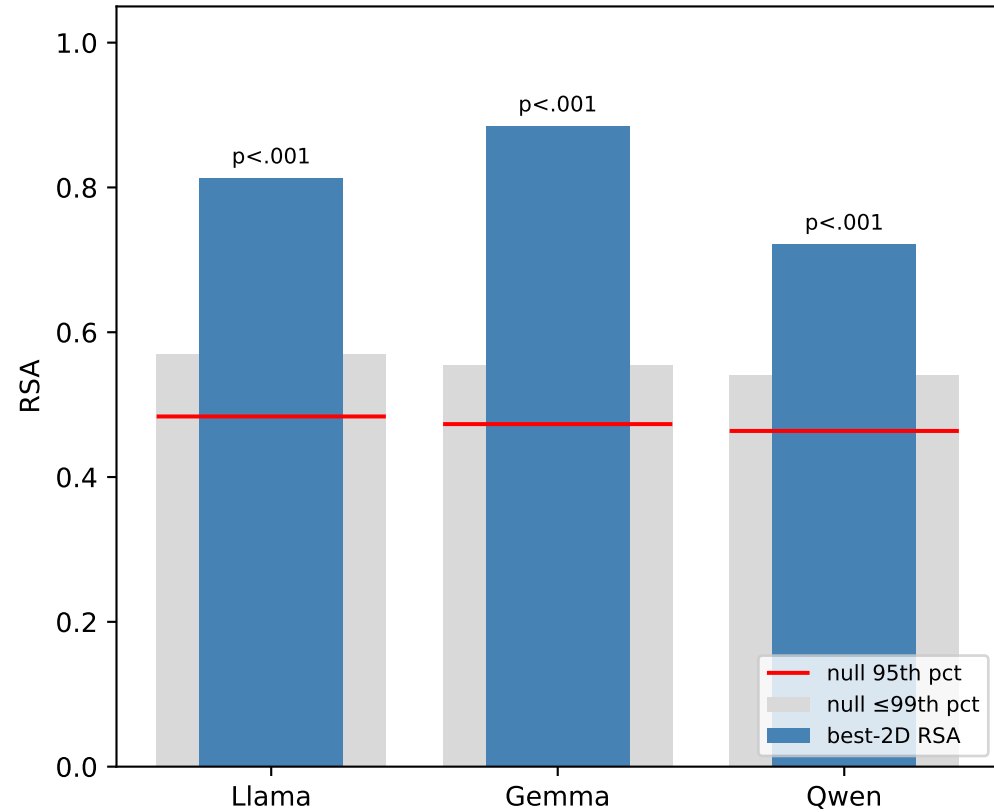


hex best-2D (RSA 0.88, $p < .001$)

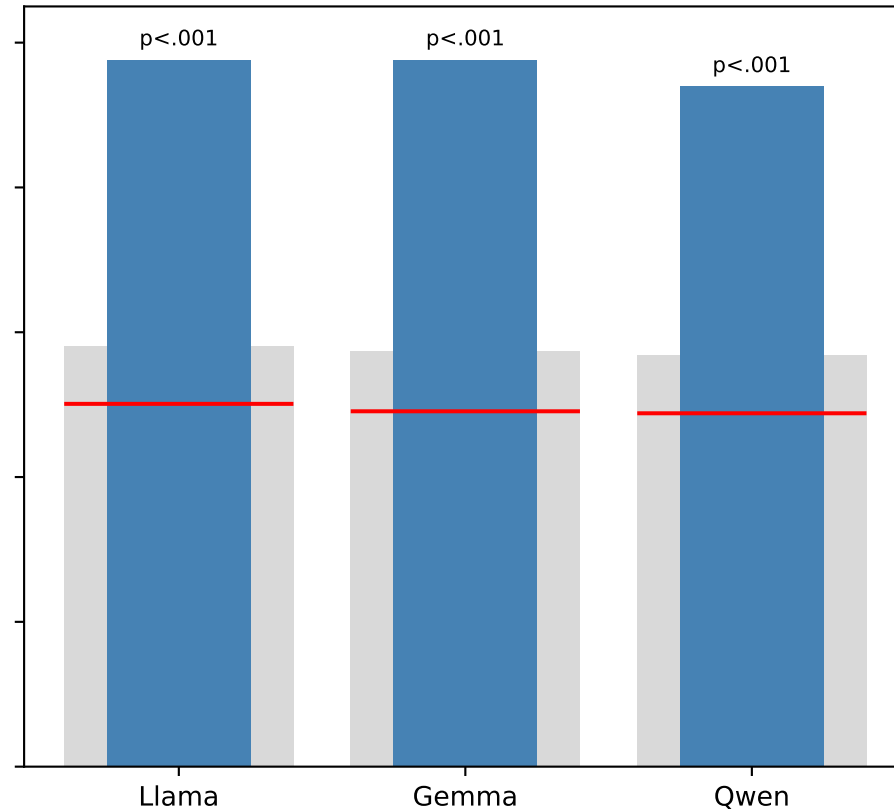


best-2D RSA vs label-permutation null (gray = null $\leq$ 99th pct, red = null 95th pct)

square_grid



ring



hex

